MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A

R/T INTERPHONE CONTROL PANEL

C-4952A/AIC-14A

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope.

This specification sheet provides performance and physical characteristics for the R/T interphone control panel C-4952A/AIC-14A, hereinafter referred to as C-4952A.

1.2 Military Part Number

The military part number shall be M2232/18-1

2. APPLICABLE DOCUMENTS

Refer to 2.2 of MIL-I-22352(AS) for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item Definition. The R/T interphone control panel shall provide the operator with the capability to activate interconnections with on-board receivers and transmitters. The operator using the C-4952A shall have the capability to turn the R/T Interplane Control panel OFF, interact with the

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associated radio transmitters and receivers only, or have access to both interphone and radio signals. The operator shall have the capability of selecting from: (1) the OFF mode, (2) a receive only mode, and (3) a receive/transmit (R/T) mode. Interphone volume control shall also be provided.

3.1.1 Item Diagram. R/T Control Panel C-4952A shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface definition. The R/T Interphone Control Panel input and output signals will normally display the characteristics presented in Table I.

TABLE I. Interface signal list.

Signal Name	Characteristics
Inputs:	
Receiver No. 1 (RCVR NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 15 Vac. No dc component normally present.
Receiver No. 2 through Receiver No. 11 (RCVR NO. X)	Same as Receiver No. 1
Interphone No. 1 (INTPH NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Interphone No. 2 through Interphone No. 10 (INTPH NO. X)	Same as Interphone No. 1
Common Audio (COMMON AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Interphone (INTPH)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac.

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TABLE I. Interface signal list - continued.

Signal Name	Characteristics
Inputs (Cont.):	
Transmit Audio (XMIT AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Emergency Receiver No. 3 (EMER RCVR NO. 3)	Relay logic signal: 1) Open 2) Short to ground
Transmit Key (XMIT KEY)	Relay logic and current sink for lamps having two states: 1) Open and 2) Short to ground
CALL, Emergency Override (CALL)	Relay logic signal: 1) Open 2) Short to ground
External Key (EXT KEY)	Relay logic signal: 1) Open 2) Short to ground
Panel Lights (LIGHTS)	0 to + 28Vdc or 0 to 6V, 400 Hz
Power (PWR)	+27.5 ± 0.5 Vdc
Outputs:	
Interphone A (INTPH A)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Interphone B (INTPH B)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Selected Receiver (SEL RCVR)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.

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TABLE I. Interface signal list - continued.

Signal Name	Characteristics
Outputs (Cont.):	
Disconnect Interphone (DISC INTPH)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Transmit Audio No. 1 (XMIT AUDIO NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Transmit Audio No. 2 through Transmit Audio No. 8 (XMIT AUDIO NO. X)	Same as (XMIT) Audio No. 1
Transmit Key No. 1 (XMIT KEY NO. 1)	Relay logic: 1) lamp supply voltage 2) short to ground
Transmit Key No. 2 through Transmit Key No. 8 (XMIT KEY NO. X)	Same as Transmit Key No. 1
Emergency Receiver No. 1 (EMER RCVR NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 15 Vac. No dc component normally present.
Emergency Receiver No. 2 (EMER RCVR NO. 2)	Same as Emergency Receiver No. 1
Interphone Control (INTPH CONTROL)	Relay logic signal: 1) Open 2) Short to ground
Key Transmit (KEY XMIT)	Relay logic signal: 1) Open 2) Short to ground

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TABLE I. Interface signal list - continued.

Signal Name	Characteristics
Outputs (Cont.):	
CALL Outgo (CALL	Relay logic signal: 1) greater than 26 Vdc 2) less than 2 Vdc
Grounds: Power (PWR GND)	Power return. See 3.4.4.2 of MIL-1-22352(AS).
System (SYS GND)	Signal return. See 3.4.4.3 of MIL-1-22352(AS).

3.1.3 Controls. The R/T Interphone Control panel shall provide the controls listed in Table II.

TABLE II. C4952A/AIC-14A controls list.

Title	Function
RAD	The C-4952A shall provide a series of eleven three position toggle switches. The switches shall provide the operator at each station with the capability to select from the switch complement available to obtain the following functions: R - In the down position the signal from the associated radio receiver shall be applied, after appropriate amplification to the operator's headset. OFF - In the center position no connection is made to the associated radio receiver.

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	R/T - In the up position the signal from the associated radio receiver shall be applied after appropriate amplification, to the operator's headset. Further, if available, the operator shall initiate keying and modulation of the associated transmitter as determined by the control settings on the master Interphone Control Panel, C-2645D.
ON-ICS-OFF	The C-4952A shall provide a three position toggle switch with the following functions: OFF - In the down position the system outputs are disabled. ICS - In the center position the incoming audio interphone signals will be inhibited, allowing only radio receiver traffic to be heard. ON - In the up position the system shall present both radio receiver and audio signals from other ICS stations.
RAD VOL	The C-4952A shall provide a control for the continuous adjustment of radio signal output level.

3.1.4 Indicators. The R/T Interphone Control Panel shall provide the indicators listed in Table III.

TABLE III. ICS Indicator List.

Title	Function
RAD Lamps	There are eight RAD lamps on the C-4952A. Lamp full on, indicates at the station that designated transmitter is being keyed. Lamp flashes with incoming receiver audio when XMTR associated with that lamp is not being keyed.
Panel Lamps	Provide panel illumination.

3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

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3.2.1.1.1 Receiver No. 1 through No. 11 (RCVR NO. X). The signals incoming from radio receivers shall be identified and assigned as a function of the aircraft in which the system is installed. A number of alternate plastic face plates are available to accommodate the various system configurations. See Supplement 1 for a listing of face plate options. The RCVR NO. X input characteristics shall be as follows:

3.2.1.1.1.1 Signal Frequency. The RCVR NO. X inputs shall accept drive signals which range from 300 to 7,000 Hz.

3.2.1.1.1.2 Signal Amplitude. The RCVR NO. X inputs shall accept drive signals which range in amplitude from 0 to 15 Vac.

3.2.1.1.3 Input Impedance. The RCVR NO. X inputs shall each present a 10,000 ohm ±10 percent impedance to the incoming signals when the R/T (up) or R (down) positions are selected on the RAD switch.

3.2.1.1.1.4 Isolation. The isolation between RCVR NO. X signals shall be at least 68 dB for signals ranging from 300 to 7,000 Hz.

3.2.1.1.2 Interphone No. 1 through Interphone No. 10 (INTPH No. X). The INTPH NO. 1 through INTPH NO. 10 input characteristics shall be as follows:

3.2.1.1.2.1 Signal Frequency. The INTPH NO. X input shall accept drive signals which range from 300 to 7,000 Hz.

3.2.1.1.2.2 Signal Amplitude. The INTPH NO. X inputs shall accept drive signals which range in amplitude from 0 to 10 Vac.

3.2.1.1.2.3 Input Impedance. The INTPH NO. X inputs shall each present a 62,000 ohm ±10 percent impedance to the incoming signals.

3.2.1.1.2.4 Isolation. The isolation between INTPH NO. X signals shall be at least 68 dB for signal ranging from 300 to 7,000 Hz.

3.2.1.1.3 Common Audio (COMMON AUDIO). The characteristics displayed by the COMMON AUDIO input shall be equivalent to those described for Interphone inputs of Section 3.2.1.1.2.

3.2.1.1.4 Interphone (INTPH). The INTPH signal shall be applied to the Transmit Key relay circuit of Figure 1. The INTPH input signal shall be connected to the INTPH output except when the Transmit Key relay is energized at which time it shall appear at the DISC INTPH output. The INTPH signal input shall have the following characteristics:

3.2.1.1.4.1 Signal Frequency. The INTPH input shall accept drive signals which range from 300 to 7000 Hz.

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3.2.1.1.4.2 Signal Amplitude. The INTPH input shall accept drive signals which range in amplitude from 0 to 5 Vac.

3.2.1.1.5 Transmit Audio (XMIT AUDIO). The XMIT AUDIO signal shall be applied to the eleven RAD switches. The XMIT AUDIO input shall display the following characteristics:

3.2.1.1.5.1 Signal Frequency. The XMIT AUDIO input shall accept drive signals which range from 300 to 7000 Hz.

3.2.1.1.5.2 Signal Amplitude. The XMIT AUDIO input shall accept drive signals which range in amplitude from 0 to 10 Vac.

3.2.1.1.6 Emergency Receiver No. 3 (EMER RCVR No. 3). The EMER RCVR NO. 3 input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The input shall be used to activate the emergency condition whereby the receiver inputs (RCVR NO. 1 through RCVR NO. 3) selected for distribution are connected to the EMER RCVR NO. 1 and EMER RCVR NO. 2 outputs. The emergency condition shall be established when the EMER RCVR NO. 3 input is an open circuit.

3.2.1.1.7 Transmit Key (XMIT KEY). The XMIT KEY input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The input shall be used to sink transmit indicator lamp current from one lamp. The lamp may be sourced from 28 Vdc or from 6 Volts at 400 Hz. In addition, the XMIT KEY signal shall be used to produce the XMIT KEY NO. X output on the line associated with the left most RAD switch selected ON.

3.2.1.1.8 CALL, Emergency Override (CALL). The CALL input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The CALL input shall be applied to the CALL Relay function which disables the interphone output of Section 3.2.1.1.4.

3.2.1.1.9 External Key (EXT KEY). The EXT KEY input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The EXT KEY signal shall be applied to the ON-ICS-OFF switch.

3.2.1.1.10 Panel Lights (LIGHTS). The C-4952A shall be capable of accepting power ranging up to +28 Vdc or 6 Vac to illuminate the three front panel lamps, one ICS OFF indicator and eight transmit indicator lamps. The panel lamps used shall be selected accordingly from TABLE IV.

TABLE IV. Front panel bulbs.

Nomenclature	Power
MS25237A-327	+28 Vdc
MS25237A-328	6 Vac

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3.2.1.1.11 Power (PWR). The C-4952A shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the R/T Interphone Control Panel shall not exceed 300 milliamperes.

3.2.1.2 Output Characteristics. All signals which are passed through this unit from input to output being routed only through connectors, wire, printed wiring, switches, and relays shall not experience insertion loss in excess of 0.5 dB. The C-4952A R/T Interphone Control Panel shall have the following output characteristics:

3.2.1.2.1 Interphone A (INTPH A). The INTPH A output shall be the same as the INTPH input except when the XMIT KEY function is activated. Then the INTPH A output shall be an open circuit. The Transmit Key relay shall be energized when the ON-ICS-OFF switch is OFF or when the Transmit Key input is a relay logic low (ground). The activation of the relay in either of these conditions shall be overridden when a relay logic low (ground) is present at the CALL input.

3.2.1.2.2 Interphone B (INTPH B). The INTPH B output shall be the sum of INTPH NO. 1 through NO. 10 and the COMMON AUDIO signals reduced 40.4 ± 1dB in amplitude from the individual input value(s) when terminated in 600 ohms. The contribution to the total by each input shall be equal ± 10 percent.

3.2.1.2.3 Selected Receiver (SEL RCVR). The SEL RCVR output shall be the sum of the operator selected RCVR NO. 1 through RCVR NO. 11 inputs adjusted (reduced) in amplitude by the RAD VOL control. Each RCVR input shall contribute equally ± 10 percent to the total output. The RAD VOL control shall demonstrate an attenuation range of 30 dB.

3.2.1.2.4 Disconnect Interphone (DISC INTPH). The DISC INTPH output shall be an open circuit except when the Transit Key relay function is activated. The state of the Transit Key relay is dependent upon three inputs: 1) CALL, 2) the ON-ICS-OFF switch, and 3) XMIT KEY.

If the C-4952A is turned OFF by placing the ON-ICS-OFF switch in the OFF position, the INTPH input shall be routed to the DISC INTPH output through the activated Transit Key relay. Also if the C-4952A receives a XMIT KEY (ground) input indicating a desire by the operator to transmit, the INTPH input shall be routed to the DISC INTPH. The activation of the Transit Key relay under these two conditions may be overridden by a CALL (ground) input which shall open the DISC INTPH output and route the INTPH input signal to the INTPH A output.

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3.2.1.2.5 Transmit Audio No. 1 through Transmit Audio No. 8 (XMIT AUDIO NO. X). Only one of the XMIT AUDIO NO. 1 through XMIT AUDIO NO. 8 outputs shall have an active signal at any one time. The XMIT AUDIO NO. X output line shall be taken from the XMIT AUDIO input. The specific active line to display the output shall be determined by operator selection of a RAD switch to the R/T position, the leftmost switch having priority.

3.2.1.2.6 Transmit Key No. 1 through Transmit Key No. 8 (XMIT KEY NO. X). The XMIT KEY No. X output shall be a relay drive logic signal having two states: 1) lamp supply voltage and 2) short to ground. The output shall be a short to ground when the associated RAD switch is the leftmost switch selected to R/T position and the XMIT KEY input is a ground.

3.2.1.2.7 Emergency Receiver No. 1 and Emergency Receiver No. 2 (EMER RCVR NO. X). The EMER RCVR NO. 1 output shall be connected to the EMER RCVR NO. 2 within the C-4952A. Therefore, the two output characteristics shall be identical. The EMER RCVR outputs shall be open circuit except when EMER RCVR NO. 3 input is a relay logic low. The EMER RCVR outputs shall be the sum of the RCVR NO. X signals associated with the three upper leftmost RAD switches.

3.2.1.2.8 Interphone Control (INTPH CONTROL). The INTPH CONTROL shall be taken from the ON-ICS-OFF switch. The INTPH CONTROL signal shall be open circuit when the ON-ICS-OFF switch is OFF. When ON is selected, the INTPH CONTROL shall be identical to the EXT KEY input.

3.2.1.2.9 Key Transmit KEY (XMIT). The KEY XMIT output shall be taken from the ON-ICS-OFF switch. When the ON-ICS-OFF switch is ON, the output shall be an open circuit. When the ON-ICS-OFF switch is OFF, the output shall be identical to the EXT KEY input.

3.2.1.2.10 CALL, Outgo (CALL). The CALL output shall be a relay status sensing line having two states: 1) HI and 2) LO where a high level output exceeds 26 Vdc and a low level output is less than 2 Vdc. The output shall respond to the state of the CALL input in accordance with the following truth table:

CALL	CALL (OUTGO)
Open	Hi
Ground	Low

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3.2.2 Physical characteristics.

3.2.2.1 Dimensions. The C-4952A shall consist of a single panel-mounted assembly having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The C-4952A shall weigh no more than 2.5 pounds.

3.3 DESIGN AND CONSTRUCTION

The design and construction of the R/T Interphone Control panel C-4952A shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

3.3.1 Plastic lighting plates. Plastic lighting plates shall meet the requirements of MIL-P-7788 and Figure 2 of this specification. Markings shown on Figure 2 shall conform with the appropriate application requirements of MS18050. A listing of lighting plates for the C-4952A is contained within Supplement 1 of MIL-I-22352(AS). If the appropriate configuration drawings are not specified in the contract, this information will be supplied by the procuring activity upon application by the contractor.

3.3.2 Front panel lighting. The front panel lamps shall be installed using mounted hardware having red filters in accordance with MIL-L-25467 to provide night vision. The panel lamps shall be mounted in accordance with the outline drawing Figure 2. The lamp drive circuitry shall be capable of accommodating power ranging up to 28 Vdc or 6 V, 400 Hz ac.

4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

4.1 Responsibility for tests and inspections. See MIL-I-22352(AS).

4.2 Test Equipment. See 4.3.3.1 MIL-1-22352(AS).

4.3 Test Methods.

4.3.1 Examination of Product. See 4.3.3.2.1 MIL-1-22352(AS).

4.3.2 Operational Test. See 4.3.3.2.2 MIL-1-22352(AS).

4.3.2.1 Input Impedance. Verify that the input impedances are in accordance with 3.2.1.1.1.3 herein.

4.3.2.2 Logic. Verify the logic requirements of 3.2.1.2.1, 3.2.1.2.2.2, 3.2.1.2.3, 3.2.1.2.4, 3.2.1.2.5, 3.2.1.2.6, 3.2.1.1.7, 3.2.1.2.8, 3.2.1.1.9, and 3.2.1.2.10.

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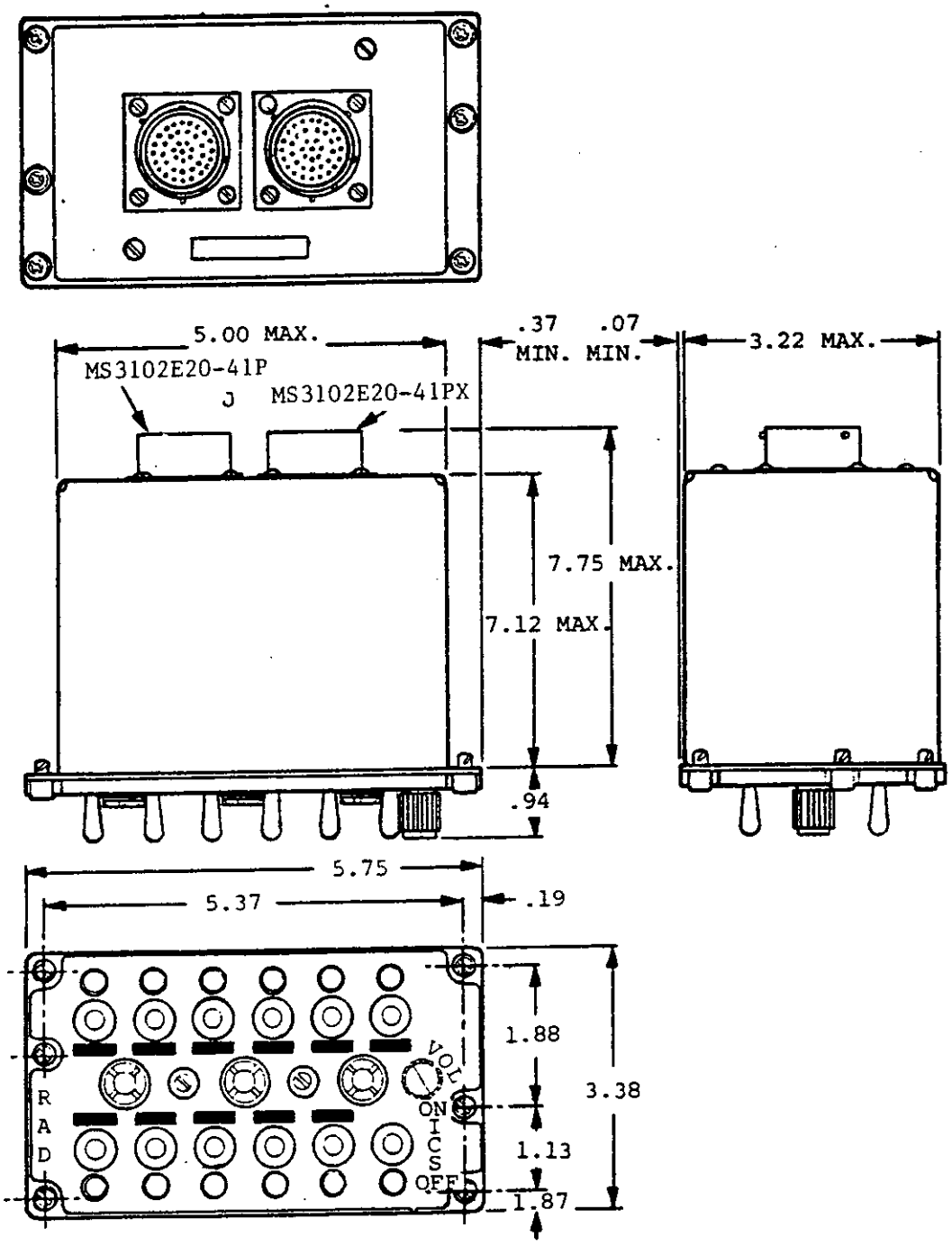
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NOTE:

1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. C-4952A outline drawing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/18(AS)	INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A R/T INTERPHONE CONTROL PANEL C-4952A/AIC-14A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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